
3AL Tutorial 5 2024 Solutions

1. Prove parts (v) and (vi) of the Correspondence Theorem: Let G be a group and $K \triangleleft G$. Define a map

$$\Theta : \{H : H \text{ is a subgroup of } G, K \subseteq H\} \rightarrow \{\mathcal{H} : \mathcal{H} \text{ is a subgroup of } G/K\}$$

by $\Theta(H) = H/K$. If H_1 and H_2 are subgroups of G containing K , then

$$(v) \quad \Theta \text{ preserves intersections: } (H_1/K) \cap (H_2/K) = (H_1 \cap H_2)/K,$$

$$(vi) \quad \Theta \text{ preserves products: if } H_1H_2 \leq G \text{ then } (H_1/K)(H_2/K) = (H_1H_2)/K.$$

(v) First, we show that $(H_1/K) \cap (H_2/K) \subseteq (H_1 \cap H_2)/K$. Take an arbitrary element $x \in (H_1/K) \cap (H_2/K)$. Since $x \in (H_1/K)$, we have $x = h_1K$ for some $h_1 \in H_1$. Similarly, $x = h_2K$ for some $h_2 \in H_2$. Therefore, $h_1K = h_2K$ and so $h_1^{-1}h_2 \in K$. Since $K \subseteq H_1$, we have $h_2 = (h_1h_1^{-1})h_2 \in H_1$. Therefore, $h_2 \in H_1 \cap H_2$. Thus $x = h_2K \in (H_1 \cap H_2)/K$.

Now let $x \in (H_1 \cap H_2)/K$. Then $x = hK$ for some $h \in H_1 \cap H_2$. Thus $x \in H_1/K$ and $x \in H_2/K$. So, $x \in (H_1/K) \cap (H_2/K)$.

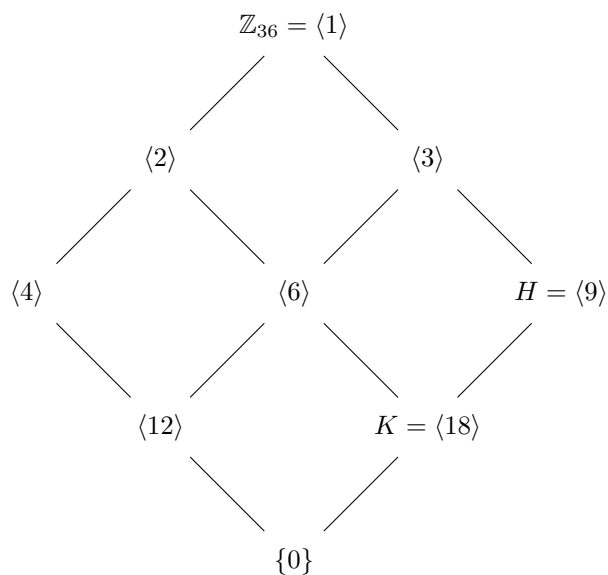
(vi) Suppose that H_1H_2 is a subgroup of G . We start by showing $(H_1/K)(H_2/K) \subseteq (H_1H_2)/K$. Take $x = h_1Kh_2K \in (H_1/K)(H_2/K)$. Therefore, $x = h_1h_2K$ and so $x \in (H_1H_2)/K$.

Now take $x \in (H_1H_2)/K$. Then $x = h_1h_2K$ for some $h_1 \in H_1$ and $h_2 \in H_2$. However, $h_1h_2K = (h_1K)(h_2K)$, where $h_1K \in H_1/K$ and $h_2K \in H_2/K$. Thus $x \in (H_1/K)(H_2/K)$.

2. Consider the group $G = \mathbb{Z}_{36}$ and subgroups $H = \langle [9] \rangle$ and $K = \langle [18] \rangle$.

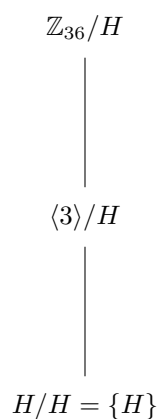
- (a) Determine all subgroups of $\mathbb{Z}_{36}$ and draw the lattice diagram.
- (b) Use the Correspondence Theorem to draw the lattice diagrams for $\mathbb{Z}_{36}/H$ and $\mathbb{Z}_{36}/K$.

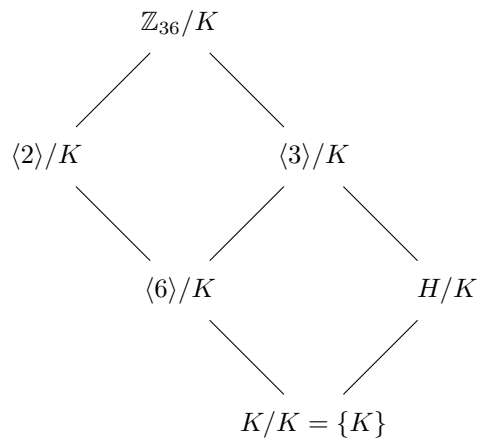
(a) Recall that the subgroups of $\mathbb{Z}_n$ are in one-to-one correspondence with the divisors of n , where each subgroup has order equal to one of the divisors. The divisors of 36 are 1, 2, 3, 4, 6, 9, 12, 18 and 36. These correspond to the cyclic subgroups $\langle[0]\rangle$, $\langle[18]\rangle$, $\langle[9]\rangle$, $\langle[6]\rangle$, $\langle[4]\rangle$, $\langle[3]\rangle$, $\langle[2]\rangle$ and $\langle[1]\rangle$. The sublattice diagram is therefore,



The equivalence class brackets have been dropped for ease of notation.

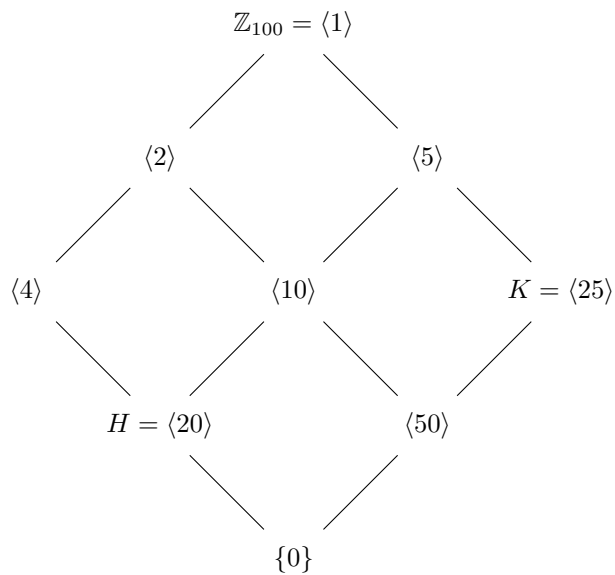
(b)



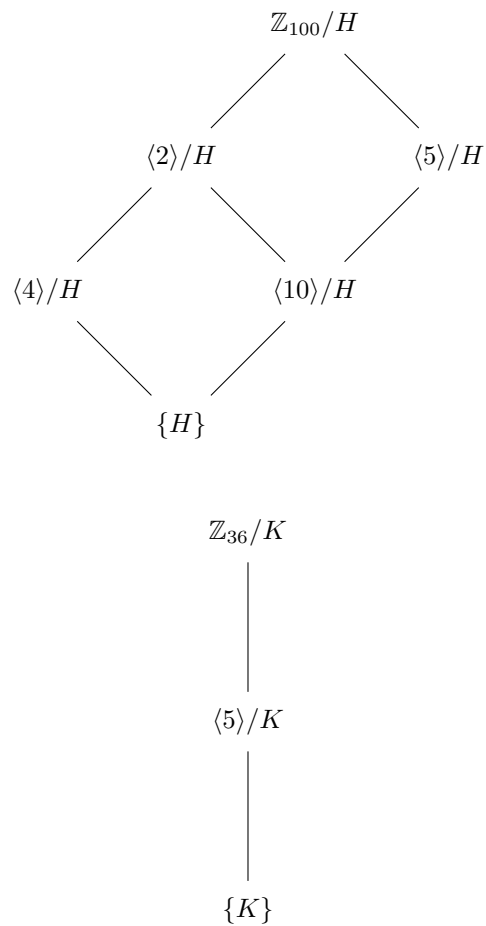


3. Repeat Question 2 with $G = \mathbb{Z}_{100}$, $H = \langle [20] \rangle$ and $K = \langle [25] \rangle$.

(a) The divisors of 100 are 1, 2, 4, 5, 10, 20, 25, 50 and 100. These correspond to the cyclic subgroups $\langle 0 \rangle$, $\langle 50 \rangle$, $\langle 25 \rangle$, $\langle 20 \rangle$, $\langle 10 \rangle$, $\langle 5 \rangle$, $\langle 4 \rangle$, $\langle 2 \rangle$ and $\langle 1 \rangle$. The sublattice diagram is therefore,



(b)



4. Let H , K and L be normal subgroups of G with $H \subseteq K \subseteq L$. Define $A = G/H$, $B = K/H$ and $C = L/H$.

(a) Show that $B \triangleleft A$, $C \triangleleft A$ and $B \triangleleft C$.

(b) What factor group is $(A/B)/(C/B)$ isomorphic to?

(a) Since H , K and L are normal in G and $H \subseteq K \subseteq L$, the Correspondence Theorem gives us $K/H \triangleleft G/H$, $L/H \triangleleft G/H$ and $K/H \triangleleft L/H$. Thus $B \triangleleft A$, $C \triangleleft A$ and $B \triangleleft C$.

(b) By the Third Isomorphism Theorem,

$$(A/B)/(C/B) \cong A/C = (G/H)/(L/H) \cong G/L.$$

5. (a) Prove that a finite group with only two subgroups (itself and the trivial group) is cyclic and of prime order.

Let's begin by proving a group with only two subgroups is cyclic and has prime order. Let G be such a group. Take a nonidentity element $g \in G$. Then $\langle g \rangle$ is a subgroup of G . Therefore, we must either have $\langle g \rangle = \{1\}$ (which is not possible since this would mean $g = 1$) or $\langle g \rangle = G$. Thus, G is cyclic.

Suppose that $|G|$ is not prime. Then there is some divisor d of $|G|$ not equal to 1 or $|G|$. Then g^d is an element of G of order $|G|/d$. Thus $\langle g^d \rangle$ is a subgroup of order $|G|/d$. Therefore the subgroup cannot be G or $\{1\}$. This contradicts the assumption that G has only two subgroups. Therefore, G must have prime order.

- (b) Let G be a finite group with $K \triangleleft G$. Show that the following conditions are equivalent:

- (i) If H is a subgroup of G with $K \subseteq H$, then $H = K$ or $H = G$,
- (ii) G/K is cyclic and has prime order.

(i) $\Rightarrow$ (ii). By the Correspondence Theorem, all subgroups of G/K must have the form H/K where H is a subgroup of G that contains K . However, then $H = K$ or $H = G$. Meaning the only subgroups of G/K are $K/K = \{K\}$ or G/K . Thus G/K is cyclic and has prime order.

(ii) $\Rightarrow$ (i). Let H be a subgroup of G with $K \subseteq H$. Then H/K is a subgroup of G/K . Therefore, its order must divide the order of G/K , which is prime, say p . Then either $|H/K| = p$ or $|H/K| = 1$. This means that either $H/K = G/K$ or $H/K = K/K$. That is, $H = G$ or $H = K$.

Extra problems

A normal subgroup K of G (where $K \neq G$) is called a **maximal normal subgroup** if the only normal subgroups H of G with $K \subseteq H$ are $H = K$ and $H = G$. A group G is called **simple** if its only normal subgroups are $\{1\}$ and G .

(I) Prove that if G is simple and $\varphi : G \rightarrow G'$ is an epimorphism, then G' is simple.

Suppose that G is simple. Let H' be a normal subgroup of G' . We will show that either $H' = \{1_{G'}\}$ or $H' = G'$. Then $H = \{g \in G : \varphi(g) \in H'\}$ is a normal subgroup of G (prove this yourself). Thus $H = \{1_G\}$ or $H = G$. If $H = \{1_G\}$, then $H' = \{1_{G'}\}$. On the other hand, if $H = G$, then, since φ is an epimorphism, we must have $H' = G'$. Thus, G' is simple.

(II) Prove that a normal subgroup K of G is a maximal normal subgroup if and only if G/K is simple.

Let K be a maximal normal subgroup of G . Let $\mathcal{H}$ be a normal subgroup of G/K . We want to prove that $\mathcal{H}$ is either $K/K = \{K\}$ or G/K . By the Correspondence Theorem, we can write $\mathcal{H} = H/K$ for some subgroup H of G with $K \subseteq H$. Again, by the Correspondence Theorem, since $H/K \triangleleft G/K$, we must have $H \triangleleft G$. Thus H is a normal subgroup of G containing K . However, K is maximal, so either $H = K$ or $H = G$. That is, $\mathcal{H} = H/K = \{K\}$ or $\mathcal{H} = H/K = G/K$. So G/K is simple.

Let K be a normal subgroup of G and suppose that G/K is simple. Suppose there is a normal subgroup H of G with $K \subseteq H$. We want to show that $H = K$ or $H = G$, that is, K is maximal. Since $H \triangleleft G$, the Correspondence Theorem says that $H/K \triangleleft G/K$. However, G/K is simple, so we must have either $H/K = K/K$ or $H/K = G/K$. By (iii) of the Correspondence Theorem, this means that $H = K$ or $H = G$. So, K is maximal.

(III) Let $K \triangleleft G$ be a maximal normal subgroup of G and $H \triangleleft G$ where $K \not\subseteq H$.

(i) Justify why $G = HK$.

HINT: Show $HK \triangleleft G$.

(ii) Show that $H/(H \cap K)$ is simple.

(i) The lemma preceding the example on metabelian groups says that if H and K are normal subgroups of G , then $HK \triangleleft G$. We have $K \subseteq HK$ but K is maximal so we must have either $HK = K$ or $HK = G$. The former is impossible since $K \not\subseteq H$ and so $HK \neq K$. So we must have $G = HK$.

(ii) By the Second Isomorphism Theorem,

$$HK/K \cong H/(H \cap K).$$

However, $HK/K = G/K$. By (b), since K is maximal, G/K must be simple. Thus $H/(H \cap K)$ is isomorphic to a simple group and so it is simple (by (a)).